

Human Computer Interaction and Design thesis on:

Onboarding for a Business-to-Business application to set up an Internet of Things device



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and Computer Science



Content

1. Background
2. Literature review
3. Methodology
4. Prototype
5. Expert evaluation
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Background

Definitions

Business-to-business

A product or service is sold by one business to another business [7]

Internet of Things device

An object connected to the internet with information sensing capabilities [12]

Onboarding

Process the first time a user interacts with a product [4].

Case company



Fintech start-up aiming to replace printed receipts with digital ones through an IoT device for businesses. The device is managed through an app.



Case company



“Redesign the onboarding screens of the B2B app to save costs on customer service and improve the user experience”

Introduction

Mobile apps are a key medium for digital interaction [1]

90%

of top 100 global brands offer at least one app



User onboarding impacts user-engagement and long-term use [4, 5]

Physical IoT device onboarding is under-explored while relevancy increases rapidly [9, 10, 11]

So there is a need for simple, streamlined and user-friendly onboarding applications [9]

How to design the onboarding screens of a B2B application to set up an IoT device?

Develop, test and refine design principles for this context

Literature Review



Semi-systematic literature review

Understand existing knowledge on onboarding design in general

Table 2.1: Keywords, search terms and results count

Keywords	Search terms	Included in search			
Autonomous setup	onboarding, guided setup, first-time user experience, device setup,				
	onboarding wizard, zero-touch setup, walkthrough UI, Configuration flow, autonomous setup	X	X	X	X
Hardware	smart home, hardware, IoT, device,				
	connected device mobile app.	X	X	X	X

Gamification	UX design					
	gamification, game mechanics, digital rewards system, interactive engagement				X	
Cost reduction	cost reduction, operational cost savings, process optimization, efficiency improvement, self-service efficiency, support cost reduction, workforce cost minimization, automation-driven savings, lean UX, ROI-driven cost analysis					X
		Results count	23	126	5	0

Total
154 papers



Title & abstract
32 papers



In-depth reading
20 papers

Semi-systematic literature review - Results

Methods

Interact with prototype (9)
Qualitative (9)
User interviews (6)
Think aloud (5)

Onboarding context

Software (7)
Hardware (5)

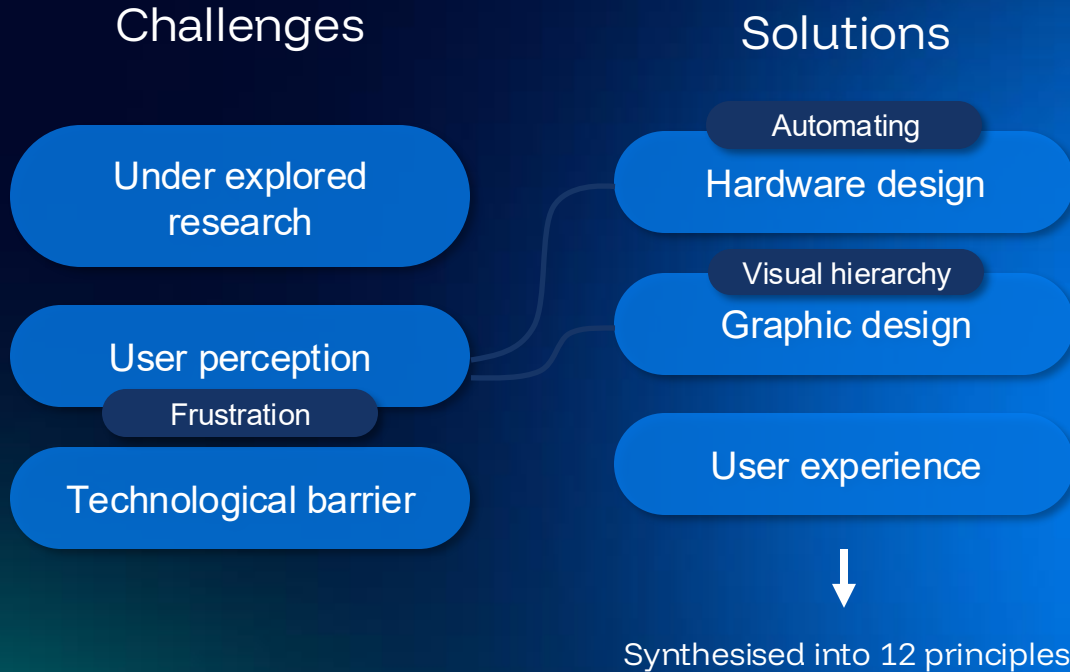
UX metrics

Task completion (4)
Understanding (3)
Perceived usability (3)
Required effort, Time to complete, Clarity, Engagement, Trust, Privacy, Navigation (2)

Users

Family (5)
Usability experts (4)
First-time users (4)
Students (2)

Semi-systematic literature review - Results



	Principle	Practical design implementation	Nielsen heuristic
1	Minimize manual effort	Automatically share Wi-Fi credentials [9] Improve pairing usability by using NFC over QR codes [9] Simplify account creation by letting users log in with Google or Apple instead of making new accounts [9]	5: Error prevention 7: Flexibility and efficiency of use
2	Optimise tech stack	Prioritise scalability by using HTML and smoothed animations using JSON [18] Ensure smooth compatibility between IoT devices [5]	4: Consistency and standards
3	Use seamless onboarding and installation flow	Identify five stages: device discovery, Wi-Fi provisioning, device pairing, registration, authorization ([9]) Minimise use of installation hardware [18]	8: Aesthetic and minimalist design
4	Reduce task flow complexity and cognitive load	Reorganise and group content [18] Break tasks down into steps [18, 31] Visualise progress with a bar [18, 25, 22] Put harder tasks at the beginning and monotonous tasks at the end [18] Ensure consistency in design, phrasing, formatting [30] Remove irrelevant content and information [31, 29] Balance amount and timing of information asked to avoid overwhelm [3] Minimise cognitive demand by limiting available options [22] and information [31]	4: Consistency and standards 1: Visibility of system status 6: Recognition rather than recall 8: Aesthetic and minimalist design
5	Present information clearly and user friendly	Friendly, simple, short and non-technical language [18, 20, 22] Avoid dense text [22, 23] Designing in scrolling page format can increase efficiency [20] Structure information according to user mental model [30] Use cues to guide attention [31]	6: Recognition rather than recall 8: Aesthetic and minimalist design
6	Use multimedia to create a multi sensory experience	Engage the senses [25, 32] Use recognisable visuals like images and icons to support information [18, 20, 4, 30, 31] Use a combination visual and auditory cues e.g. error sound [30] to improve accessibility [31] Use both words and imagery close together to transfer information better [31]	4: Consistency and standards 6: Recognition rather than recall 8: Aesthetic and minimalist design
7	Support first-time users	Use onboarding tutorials [20, 31] such as through guided exploration [24], step-by step [20, 30] and intuitive [22] guidance while explaining benefits [29] Use empty states and pre-filled templates [29]	5: Error prevention 6: Recognition rather than recall 7: Flexibility and efficiency of use 10: Help and documentation
8	Personalised content	Present a problem statement and value which users relate to [29] Personal welcome message [29] Use a conversational style as opposed to formal [31] Provide users with a memorabilia to remember the experience [32] Implement a call to action [29] Ensure the user expectations are aligned with reality [28, 24]	2: Match between the system and the real world
9	Adaptive interface and experience design	Know the users behaviour [25], needs, goals [28], routines [5] and knowledge [21] and design information and the UI around these [28, 25, 5, 21]. Design for an early aha moment [19] where the user realises the value of the software [29, 28] Ensure every point of interaction helps the user [28] Adapt to different stakeholders needs [29] Progressively show contextual information e.g. new features [20] for long-term retention [29]	3: User control and freedom 7: Flexibility and efficiency of use
10	Promote interactivity	Use interactive communication [25, 4] Interactive guidance [29, 4]	7: Flexibility and efficiency of use
11	Enforce positive emotions	Design according to a theme [32] Remove distractions and irrelevant cues [32, 30] Create a feeling of ownership like assembling or naming the device [18] Start with showing benefits for user instead of features [28] Create an emotional connection by designing for personal aspirations [28] Create happiness by satisfying individual and group needs by facilitating shared dialogue and memories [25]	7: Flexibility and efficiency of use 8: Aesthetic and minimalist design
12	Promote self sufficiency	Help users understand, think logically, improvise, plan and act [24] Support learning activities [24] and retention through repetition, structure, consistent and visual [30, 31] Walk through terms and conditions step by step to explain data gathered and use [23] Ensure error recoverability e.g. able to go back [22]	3: User control and freedom 5: Error prevention 9: Help users recognize, diagnose, and recover from errors

Table 2.7: Literature design principles, practical implementation and heuristic validation

implementation	Nielsen heuristic
Share Wi-Fi credentials [9] Improve pairing usability by using NFC over QR Facilitate account creation by letting users log in with Google or Apple instead of accounts [9] Facilitate pairing by using HTML and smoothed animations using JSON [18] Ensure smooth interaction between IoT devices [5] Guidelines: device discovery, Wi-Fi provisioning, device pairing, registration, and updates 9)) Minimise use of installation hardware [18] Organise group content [18] Break tasks down into steps [18, 31] Visualise progress with a progress bar Put harder tasks at the beginning and monotonous tasks at the end [18] Ensure consistency in design, phrasing, formatting [30] Remove irrelevant content and information Control the amount and timing of information asked to avoid overwhelm [3] Minimise cognitive load by limiting available options [22] and information [31] Use simple, short and non-technical language [18, 20, 22] Avoid dense text [22, 23] Control page format can increase efficiency [20] Structure information according to a model [30] Use cues to guide attention [31] Use visual cues [25, 32] Use recognisable visuals like images and icons to support information Use a combination visual and auditory cues e.g. error sound [30] to improve usability Use both words and imagery close together to transfer information better [31] Use guided exploration [20, 31] such as through guided exploration [24], step-by step [20, 30] Use visual guidance while explaining benefits [29] Use empty states and pre-filled states Personalise the welcome statement and value which users relate to [29] Personal welcome message [29] Use an informal style as opposed to formal [31] Provide users with a memorabilia to enhance their experience [32] Implement a call to action [29] Ensure the user expectations are met [28, 24] Understand user behaviour [25], needs, goals [28], routines [5] and knowledge [21] and design the UI around these [28, 25, 5, 21]. Design for an early aha moment [19] where the user realises the value of the software [29, 28] Ensure every point of interaction helps the user achieve one of their different stakeholders needs [29] Progressively show contextual information [20] for long-term retention [29] Facilitate communication [25, 4] Interactive guidance [29, 4] Relate the design to a theme [32] Remove distractions and irrelevant cues [32, 30] Create a sense of ownership like assembling or naming the device [18] Start with showing benefits for the user [28] Create an emotional connection by designing for personalisation [28] Create happiness by satisfying individual and group needs by facilitating shared experiences [25] Understand, think logically, improvise, plan and act [24] Support learning activities through repetition, structure, consistent and visual [30, 31] Walk through the process step by step to explain data gathered and use [23] Ensure error handling is clear e.g. able to go back [22]	5: Error prevention 7: Flexibility and efficiency of use 4: Consistency and standards 8: Aesthetic and minimalist design 4: Consistency and standards 1: Visibility of system status 6: Recognition rather than recall 8: Aesthetic and minimalist design 6: Recognition rather than recall 8: Aesthetic and minimalist design 4: Consistency and standards 6: Recognition rather than recall 8: Aesthetic and minimalist design 5: Error prevention 6: Recognition rather than recall 7: Flexibility and efficiency of use 10: Help and documentation 2: Match between the system and the real world 3: User control and freedom 7: Flexibility and efficiency of use 7: Flexibility and efficiency of use 7: Flexibility and efficiency of use 8: Aesthetic and minimalist design 3: User control and freedom 5: Error prevention 9: Help users recognize, diagnose, and recover from errors

Validation

Foundational usability evaluation framework by Jakob Nielsen (1994).

Defines 10 principles for good interface design to identify usability issues and guide improvements

Business-to-business research

Underexplored context

B2B and IoT onboarding rarely studied compared to B2C [6, 10, 11]

Due to custom and technically complex integrations [6]

Challenge to find representative users [6]

Platform adoption and engagement drivers

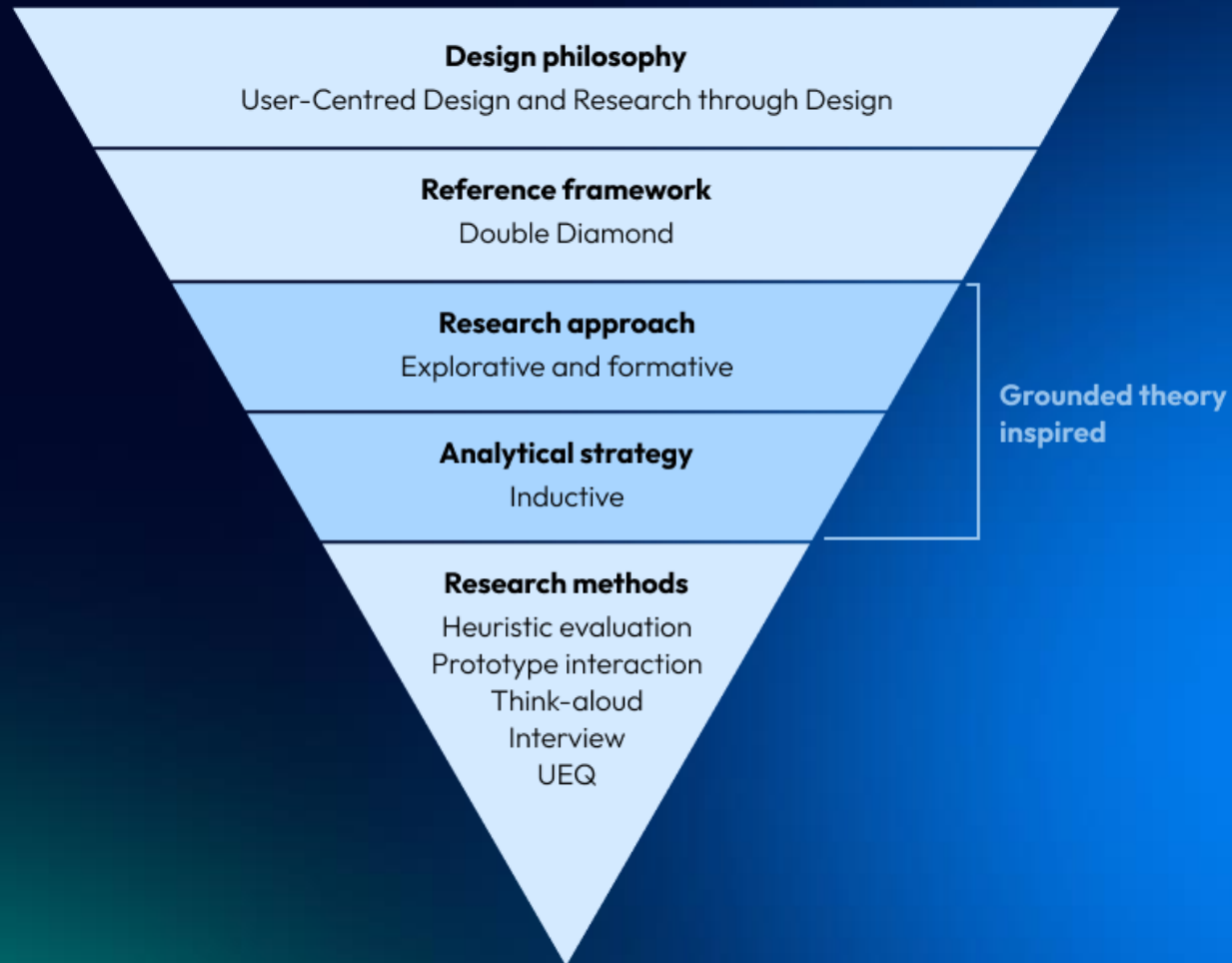
Influenced by perceived usefulness, management support and competitive pressure [37, 36]

Depends on co-creation and actor relationship [6, 40]

Usability and aesthetics play supporting role [37, 44]

Methodology





Reference framework

Double Diamond

Research approach

Explorative and formative

Analytical strategy

Inductive

Research methods

Heuristic evaluation
Prototype interaction
Think-aloud
Interview
UEQ

■ Understand domain

■ Iteratively refine design

■ Derive themes from data

User-Centred Design and Research through Design

Reference framework

Double Diamond

Research approach

Explorative and formative

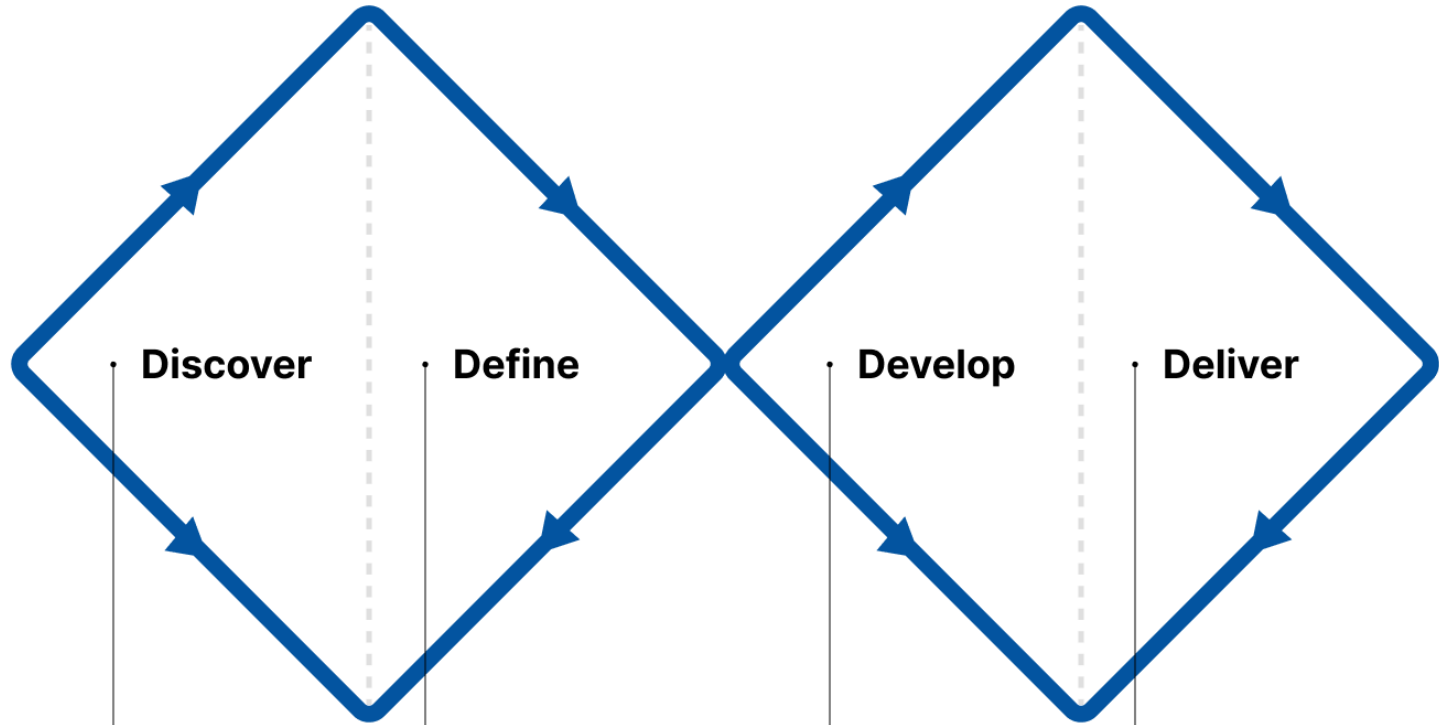
Analytical strategy

Inductive

Research methods

Heuristic evaluation
Prototype interaction
Think-aloud
Interview
UEQ

**Grounded theory
inspired**



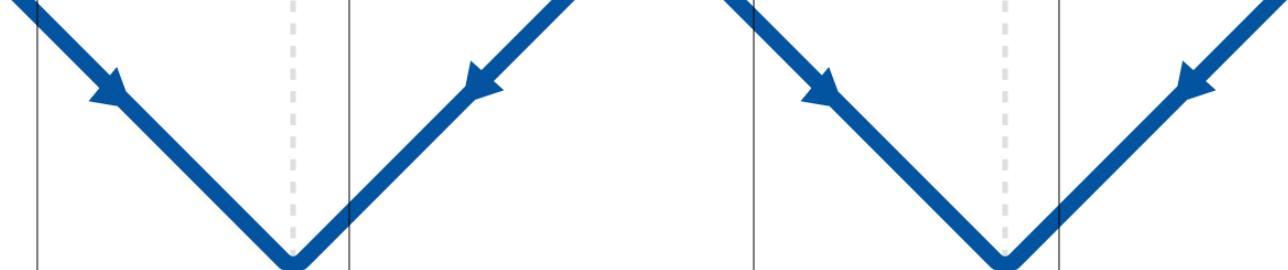
**1st
cycle**

- Semi-structured literature review

- Persona
- User journey
- Synthesise design principles

- Implement literature design principles in prototype

- Evaluate prototype with expert review heuristic evaluation



1st cycle

- | | | | |
|---|---|---|--|
| <ul style="list-style-type: none">• Semi-structured literature review | <ul style="list-style-type: none">• Persona• User journey• Synthesise design principles | <ul style="list-style-type: none">• Implement literature design principles in prototype | <ul style="list-style-type: none">• Evaluate prototype with expert review heuristic evaluation |
|---|---|---|--|

2nd cycle

- | | | | |
|---|--|--|---|
| <ul style="list-style-type: none">• Analyse expert review | <ul style="list-style-type: none">• Define relevant implementations of expert review | <ul style="list-style-type: none">• Implement expert review feedback | <ul style="list-style-type: none">• Evaluate with users |
|---|--|--|---|

3rd cycle

- | | | | |
|--|---|---|--|
| <ul style="list-style-type: none">• Thematic analysis of user evaluation | <ul style="list-style-type: none">• Compare and contrast findings with literature findings• Discuss and develop hypothesis | <ul style="list-style-type: none">• <i>Future research</i>
<i>Implement hypothesis in new context</i> | <ul style="list-style-type: none">• <i>Future research</i>
<i>Evaluation</i> |
|--|---|---|--|

Prototype





Anthony (46)

Manager of GoCeries
Supermarkets in Verona

Tagline

"As manager I focus on increasing sales revenue while improving the customer experience." [60]

Experience level

Antonio has a moderate level of digital literacy because he uses business software to reduce manual labour and increase automation [62]. However, he notices the market is increasingly offering useful smart devices but he is hesitant to adopt them due to the price and fear of struggling to use it [60, 62]. Generally he knows investing in technology would benefit his company [61], he faces challenges to adopt the fast changing technologies [60]

Context of use

Antonio manages multiple stores in different locations and often keeps an eye on sales through his laptop and phone on the go. He interacts with technology and physical devices, like receipt printers and security devices, out of necessity to support business operations. He needs to understand the systems to make sure they fit the needs of the business and serve the customer.

Goals

Improve efficiency of operations by decreasing manual labour and increasing automation while keeping things easy to use and customer friendly [62, 60]

Concerns

Price, speed and ease of use of new technology [60]

“

I don't need more overcomplicated systems to manage, I need something that just works, fast.

”

Scenario

Antonio has decided to replace his receipt printing machine with Billd's device, he has just received it in the mail and will use it for the first time. In preparation he has already downloaded the app on his phone. He will now interact with the app for the first time to set up the Billd.

Expectation

- Easy to set-up plug and play device
- Easy to understand data insights in purchases

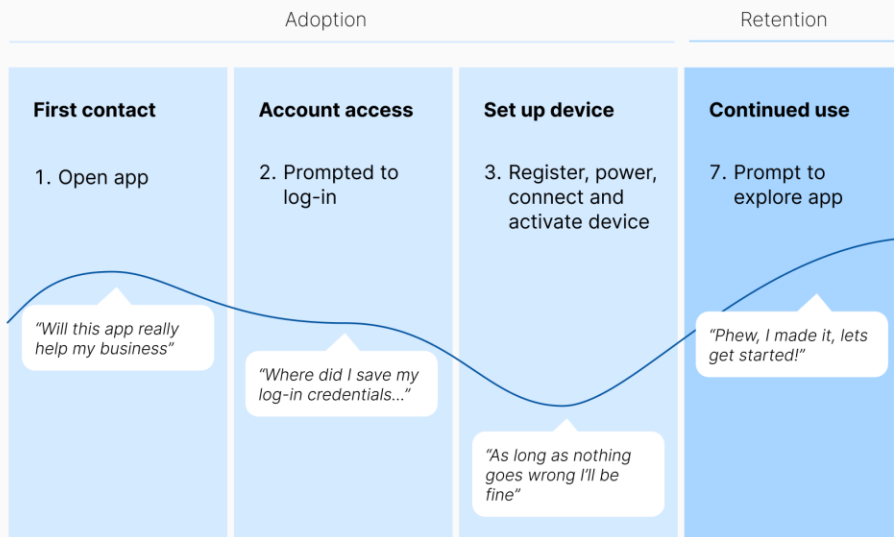
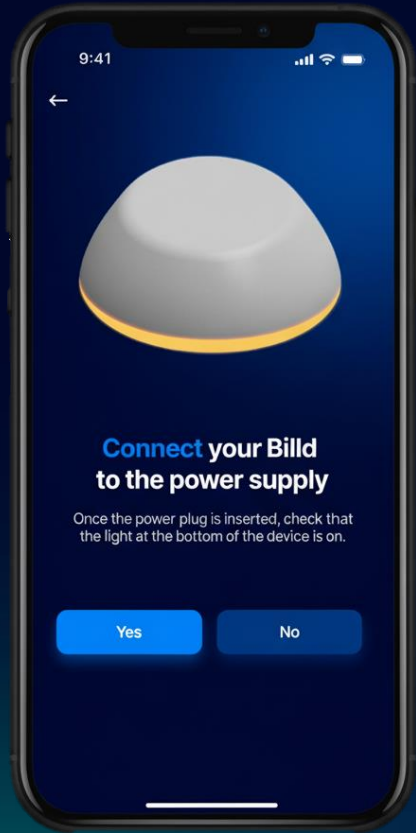


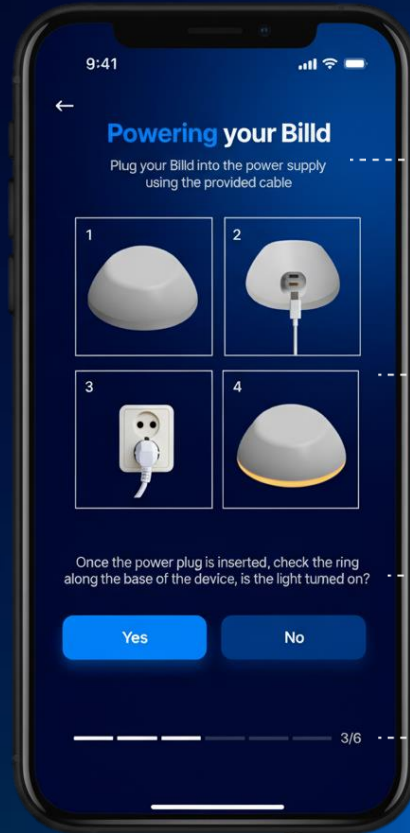
Table 4.1: Design implementation of literature principles

#	Principle	Practical design implementation	Implementation
1	Minimise manual effort	Automatically share Wi-Fi credentials [9]	TECH
		Improve pairing usability by using NFC over QR codes [9]	TECH
		Simplify account creation by letting users log in with Google or Apple instead of making new accounts [9]	TECH
2	Optimise tech stack	Prioritise scalability by using HTML and smooth animations using JSON [18]	TECH
		Ensure smooth compatibility between IoT devices [5]	IR: only using one device
3	Use seamless onboarding and installation flow	Identify five stages: device discovery, Wi-Fi provisioning, device pairing, registration and authorisation [9]	Clarity of phases improved through a progress bar
		Minimise use of installation hardware [18]	No other device is needed beyond the Billd product
4	Reduce task flow complexity and cognitive load	Reorganise and group content. [18]	Content grouped
		Break tasks down into steps [31] [18]	Added numbered steps
		Visualise progress with a bar [18, 25, 22]	Added a progress bar from step 1 to 6
		Put harder tasks at the beginning and monotonous tasks at the end [18]	IR: order of tasks is set, i.e. first logging in (monotonous), then setting up the device (complicated)
		Ensure consistency in design, phrasing and formatting [30]	Followed brand styling
		Remove irrelevant content and information [29, 31] [18]	Limit redundant information

	experience	Use recognisable visuals like images and icons to support information [18, 20, 4, 30, 31]	Designed a mascot, improved the use of images
		Use a combination of visual and auditory cues, e.g. error sound [30], to improve accessibility [31]	TECH: due to the prototype, sound was not possible, smell is irrelevant
		Use both words and imagery close together to transfer information better [31]	Used words and images close to each other
7	Support first-time users	Use onboarding tutorials [20, 31], such as through guided exploration [24], step-by-step [20, 30] and intuitive [22] guidance while explaining benefits [29]	Explained separate steps using visuals and benefits on entry screens
		Use empty states and pre-filled templates [29]	Implemented empty states in login
8	Personalised content	Present a problem statement and value which users relate to [29]	The first screen presents value
		Personal welcome message [29]	Loading screens welcome the user
		Use a conversational style as opposed to formal [31]	Used conversational style
		Provide users with memorabilia to remember the experience [32]	IR: due to it being a digital product and prototype limitations
		Implement a call to action [29]	Clear CTA at every step
		Ensure the user expectations are aligned with reality [28, 24]	Set expectations before different stages



Before



After

Friendly, simple, short and non-technical language [18, 20, 22]

Avoid dense text [22, 23]

Use onboarding tutorials [20, 31] such as through "guided exploration" [24], step-by step [20, 30] and intuitive [20]

Help users understand, think logically, improvise, plan and act [24]

Ensure the user expectations are aligned with reality [28, 24]

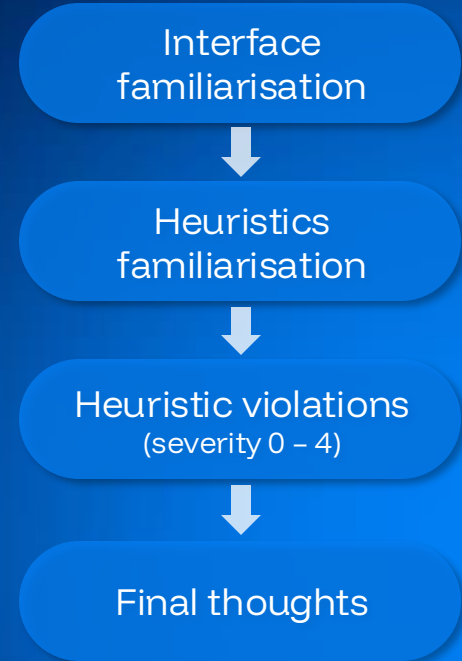
Visualise progress with a bar [18, 25, 22]

Expert Evaluation

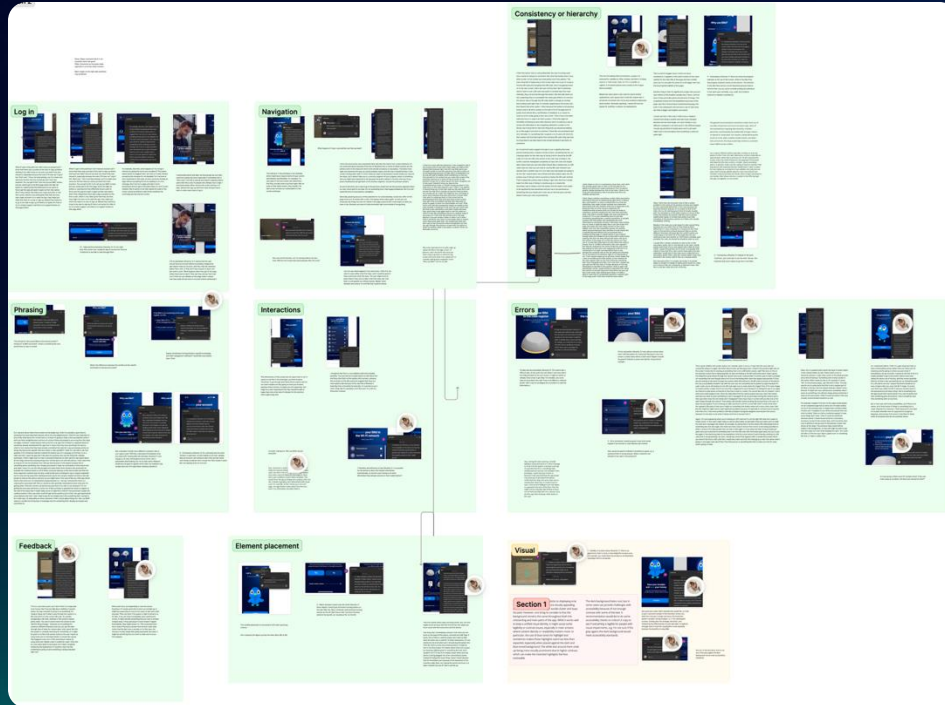


Study set up

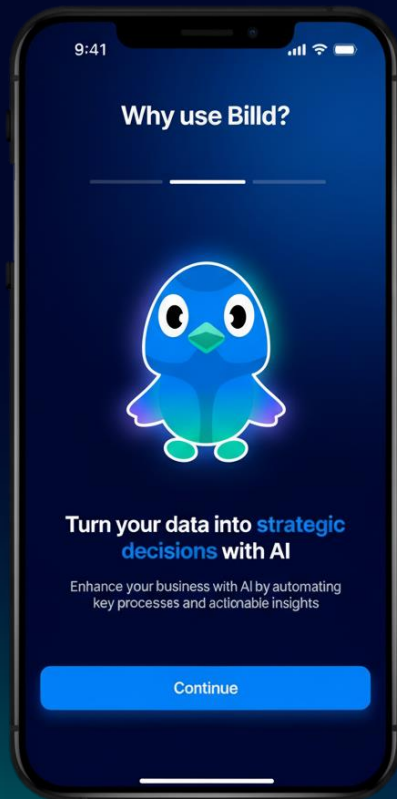
- Two 'double experts'
 - UI/UX design + B2B
 - Increased effectiveness: 81% - 90% of problems
- 45 minute independent **heuristic evaluation** expert review in Figma



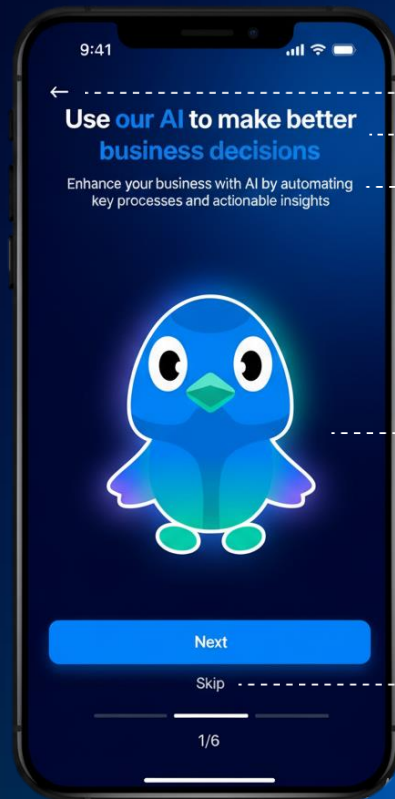
Affinity diagram analysis and results



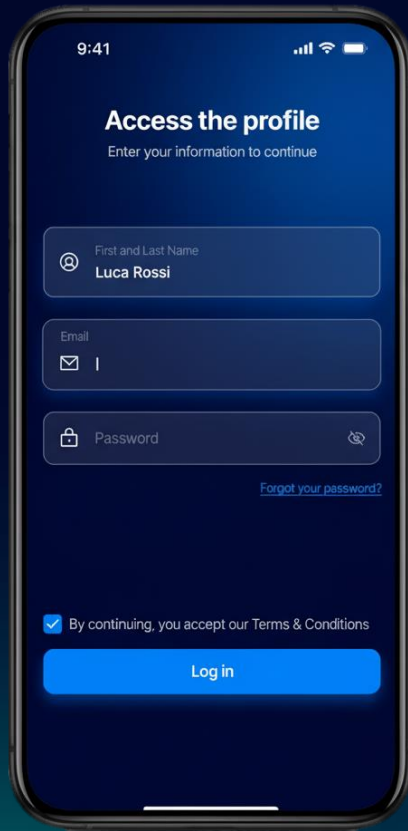
1. Log-in
2. Navigation
3. Consistency and hierarchy
4. Phrasing
5. Interactions
6. Errors
7. Feedback
8. Element placement
9. Visuals



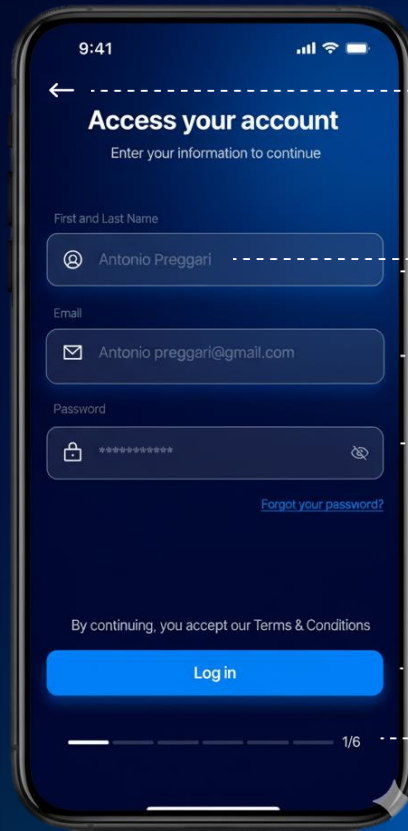
Before



After



Before



After

Ensure error recoverability e.g. able to go back [22]

Use empty states and pre-filled templates [29]

Ensure consistency in design, phrasing, formatting [30]

Use cues to guide attention [31]

Visualise progress with a bar [18, 25, 22]

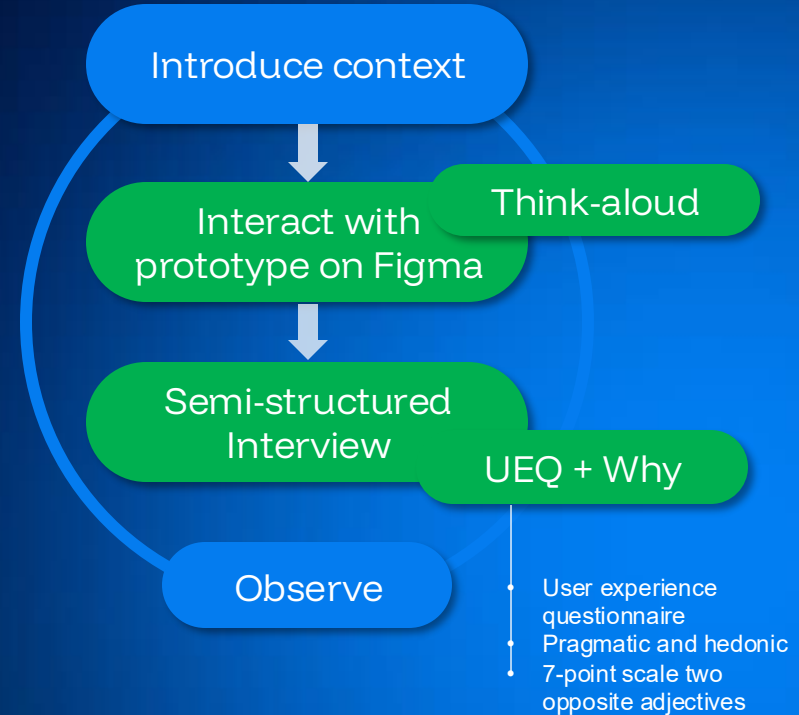
User Evaluation



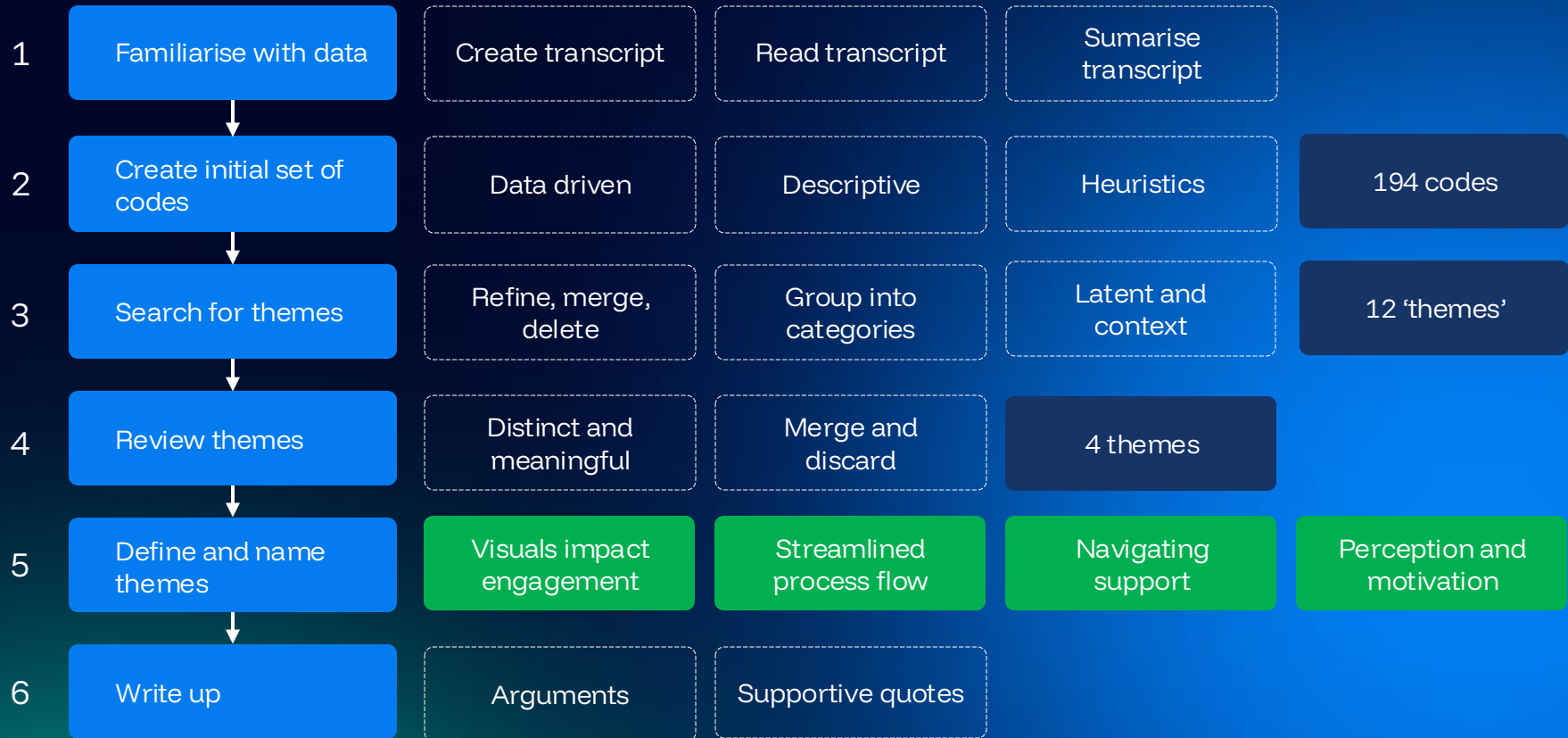
Study set up

6 participants

- Represent the end-users knowledge and experience
- Store, restaurant, operational and sustainability managers
- 25-60 years old
- Conducted via Teams with screenshare



Thematic analysis – Braun and Clarke



Visuals impact engagement

Attractive visuals evoke positive emotions

"I love the colours and everything you guys used. (...) I felt like it was exciting."

Visuals guide behaviour and actions

"Well, the button kind of lights up (...) My eyes are drawn to this button, so I say 'start now.'"

Mascot supports engagement if fitting

"I think the bird is nice, it's cheerful, it has a bit of a smile. It helps you get through it, (...)."

Streamlined process flow

Prioritise short, simple and easy language

"It looks very simple, so to say, not a lot of information is put on a small screen. It is clear."

"[There is] a lot of information with a lot more difficult words than I, as a hospitality manager, would understand (...) I would read this sentence 4 or 5 times (...) the sentence is too complex for me."

Users value speed, simplicity and step-by-step instructions

"It was easy to click through. I never want to spend so much time."

"I like that it shows you step-by-step what might be wrong."

Navigating support

Support options must be clear and obvious

“ But perhaps if I press no I’ll get a pop-up (...) [Presses ‘no’] Well, yes, perfect, that’s what I want to know.”

Help should be easy to find and match the user’s problem

“Look, this does give me insight, (...) you know someone is going to help me (...) But this text could be a little easier for me, or perhaps step-by-step easier.”

Expectation management reduces support request

“ I would mention something about that in the communication beforehand. A precondition to using this app is that you have a cash system where you can select a printer and select it based on an IP address.”

Perception and motivation

Highlighting the system value in accordance with user value

“It’s also good to explain it again, or show a customer again why they’re buying the app to remind them about the improved sales data. That’s motivating! (...) When you install an app, you don’t want to read a whole book! It should be quick and easy.”

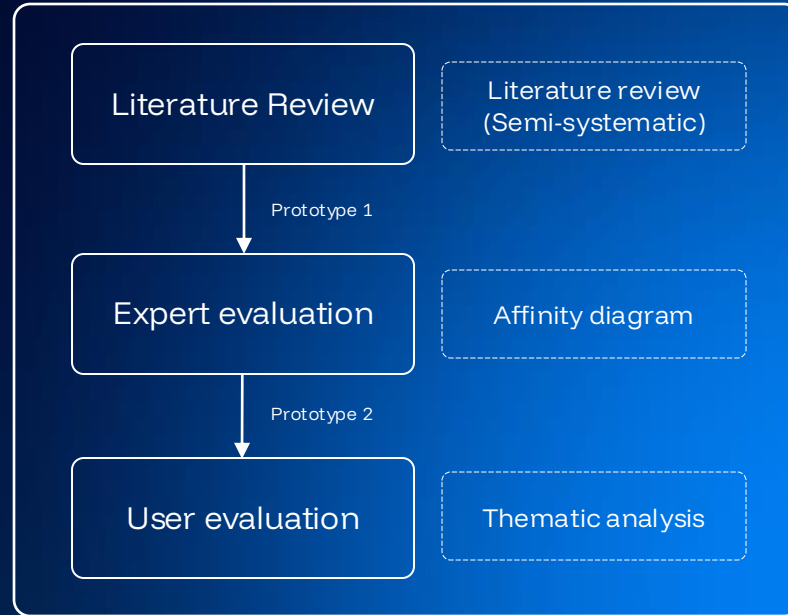
Users have varying technological experience requiring simple communication

“I sometimes think the language is a bit complex, that you already have to be comfortable with certain ICT powerful matter to do such a setting properly.”

Discussion

○ ○ ○

Interpret results



Cross validation

Table 7.1: User evaluation themes, corresponding arguments and supportive literature principle

Theme	User evaluation arguments	Supportive literature principle
1	Visuals can be used to guide the user's attention to inform user actions during onboarding.	6. Use multimedia to create a multi sensory experience
1	A visually attractive and consistent interface leads to positive emotional response of the user, thus improving engagement during onboarding.	6. Use multimedia to create a multi sensory experience 11. Enforce positive emotion
1	Mascots have the potential to support the user through the onboarding process to improve both engagement and the usability. However, effectiveness depends on the mascot's perceived fit with the application, as it may also lead to confusion and lack of coherence.	6. Use multimedia to create a multi sensory experience
2	Information is understood when presented in a short and simple manner, avoiding complex language.	5. Present information clearly and user friendly 4. Reduce task flow complexity and cognitive load
2	Users value speed, simplicity and ease when engaging in an onboarding process. This is ultimately done by reducing the cognitive load by minimizing distraction and providing clear step-by-step instructions.	5. Present information clearly and user friendly 4. Reduce task flow complexity and cognitive load

3	Support options need to be obvious, explicit and straightforward in their design, ensuring clear affordance. Poorly designed support options that are implicit, suggestive or lack affordance, can be missed or lead to confusion and reduced efficiency. Ultimately, users may find other support means and abandon the application dissatisfied.	5 Present information clearly and user friendly 4 Reduce task flow complexity and cognitive load 12 Promote Self sufficiency
3	Help offered should support the problem the user is facing and be communicated in a simple and straightforward manner. Failing to do so may lead to unnecessary trial and error, abandonment or negative emotions.	4. Reduce task flow complexity and cognitive load 5. Present information clearly and user friendly 9. Adaptive interface and experience design 12. Promote self sufficiency
3	Upfront expectation management can prevent the need for support or reduce support requests throughout the onboarding process.	7. Support first-time users 8. Personalised content 12. Promote self sufficiency
4	Aligning and highlighting the system's potential value for the user can set the stage for a positive experience. However, to enhance positive response this must be communicated mindfully, by adhering to user values of simplicity and speed.	8. Personalised content 11. Enforce positive emotions
4	Differing degrees of users' experience with technology and relevant systems can influence their confidence and enthusiasm in using the application and seeing its value. This suggests a need for material that is adapted to a wide range of technical abilities, prioritizing easy to understand language for those less experienced.	5. Present information clearly and user friendly 9. Adaptive interface and experience design

Synthesis

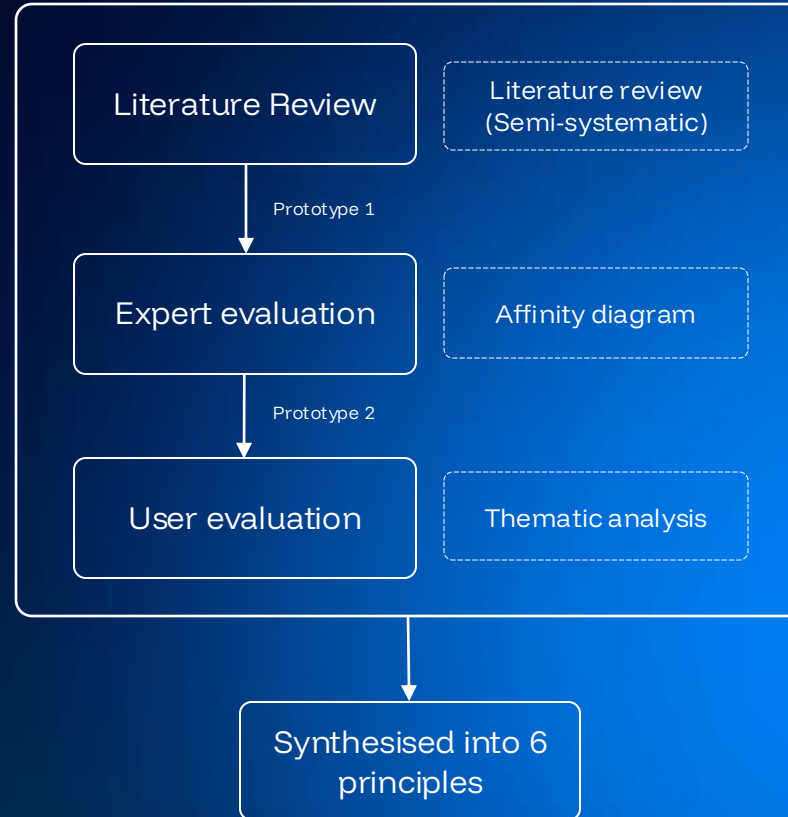


Table 7.2: Synthesised principles from literature and evaluation

	Principle	Practical design implementation
1	Reduce task flow complexity and cognitive load	Break tasks down into steps [18, 31] Limit available options [22] Aligning with the users mental model [30] Enable users to go through the process quickly by ensuring swift navigation Provide step-by-step task instructions [20, 30] to increase user control and predictability
2	Present information clearly and user friendly	Present information clearly and user friendly [22, 20] Use short and simple, non-technical language and avoiding dense text [18, 20, 22, 23] to maintain user attention and improve understanding both in task descriptions and help support Clear attention guiding cues [31] Remove irrelevant information [31, 29] Visualise progress with a bar [18, 25, 22] Minimise on-screen details to avoid distraction [32, 30]
3	Use multimedia to create a multi sensory experience	Use recognisable imagery to support information [18, 20, 4, 30, 31] Use an aesthetic colour palette to improve satisfaction Use distinctive colours to guide attention and inform actions [31] Use only realistic, accurate and precise images to inform interactions Characters, like mascots, can be used to guide attention given that they are relevant to the information and application to avoid confusion. Design according to a theme [32] Using suitable and an aesthetic colour palette improves engagement

		Present a problem statement and value which users relate [29] Adjust information to the experience level of the user [21], often meaning simplicity is key Offer skip options to account for the users priorities of speed and simplicity Show benefits for the user as opposed to features [28] Align with users personal aspirations [28]
5	Adaptive interface and experience design	Positive impact of using recognisable imagery to support information [18, 20, 4, 30, 31] Understand and design for the users behaviour, needs, goals, routines and knowledge [25, 28, 5, 21] Help the user realise the value of the software (the aha moment) [19, 29, 28] Know the user and design around them [25, 28, 5, 21] Help support should match the users level of experience and way of working
6	Promote self sufficiency	Help the user understand, think logically, improvise, plan and act [24] Help users plan ahead [24] Help buttons need to be recognisable, ensuring clear buttons affordance to help users feel supported Prioritise actionable, short and simple support Set user expectations up front, such as communicating required material Novel technologies may require additional explanation for users to see its value

Limitations

- Small-scale qualitative study
- Lacking physical prototype
- Limited to device setup instead of app usage

Future research

- Validate principles quantitatively
- Explore role of IoT device
- Test across other B2B industries
- Study long-term engagement

Literature gave insight on

Designing onboarding flows for a B2C predominantly software context.

The knowledge gap was

Onboarding in a B2B context that involves setting up an IoT device. While evaluation with real B2B end-users.

This study contributes

Academically

Extends onboarding research in B2B IoT context, revealing engagement and usability factors.

Practically

Delivers six design principles guiding effective, user-friendly B2B IoT onboarding flows.

Methodologically

Triangulated exploratory, formative, and inductive approaches within Double Diamond framework.

Thank you!

Questions?

